Garvit Goyal

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Education

Georgia Institute of Technology December 2022
M.S. in Computational Science and Engineering Atlanta, GA

Indian Institute of Technology Bombay (IIT Bombay) August 2020
B.Tech. in Civil Engineering Mumbai, India
Thesis: Machine Learning for Vehicle Speed and Lateral Position Prediction on Rural Highway Curves
Advisor: Avijit Maji

Research Interests

Reliable and Trustworthy Machine Learning: Mechanistic Interpretability, Reliability, and Robustness.
Meta-learning and Automated Machine Learning: Hyperparameter Optimization, Algorithm Selection, and Dataset Meta-features.
Graph and Geometric Representation Learning: Graph Neural Networks and Non-Euclidean Representations.

Research Experience

Independent Researcher May 2026–Present

- Designed a layerwise study of high-confidence errors in tabular foundation models, TabPFN, TabICL, and TabDPT, including the pipeline needed to extract and compare their internal representations.
- Designed and evaluated reliability measures based on class support, nearest-neighbor structure, and class geometry, using confidence matching, permutation tests, and cross-model controls.

IIT Bombay March 2026–Present
Research Collaborator, with Avijit Maji

- Formulated tuning gain and optimizer sensitivity as separate pre-HPO prediction problems and developed a data-complexity-based approach for studying both.
- Designed and implemented the experimental pipeline, including optimizer benchmarking, complexity analysis, statistical comparisons, robustness tests, and mixed-variable exploratory landscape analysis.
- Led the analysis, interpretation, and manuscript preparation for journal submission.

IIT Bombay May 2017–July 2020
Undergraduate Researcher, with Avijit Maji and Anuj Budhkar

- Developed a trajectory-processing pipeline to identify inter-vehicle interactions from roadway video and incorporate them into Random Forest models of vehicle speed and lateral position on highway curves.
- Analyzed how roadway geometry, trajectory history, and interaction context affected predictions at different locations along the curve; this work formed the basis of the undergraduate thesis.
- Designed and calibrated microsimulation and driving-simulator experiments to study merging and post-crash behavior in mixed traffic, including shockwave, headway, and platoon-dispersion analyses.

Publications & Manuscripts

Garvit Goyal and Avijit Maji. “Before You Tune: Data Complexity Metrics as Indicators of Hyperparameter Optimization Outcomes” *Under review at TMLR*.
Garvit Goyal. “Data Complexity and Selective Reliability in Tabular Foundation Models” *Springer Proceedings in Mathematics & Statistics*, Springer, in press, 2026.

Projects

Geometric and Neural Pooling Methods for Sentence-BERT 2022
Georgia Institute of Technology

- Developed neural and geometric alternatives to mean pooling in Sentence-BERT, including convolutional pooling, Lorentzian sentence embeddings, and alignment–uniformity objectives.
- Implemented and compared Euclidean and hyperbolic variants on semantic textual similarity benchmarks; convolutional pooling substantially improved sentence representations, while the non-Euclidean formulations revealed optimization and numerical-stability constraints.

Adaptive Dual-Branch Graph Convolutional Networks for Imbalanced Node Classification 2022

Georgia Institute of Technology

- Developed A-DGCN, a dual-branch graph convolutional architecture that replaces k-nearest-neighbor graph reconstruction with class-frequency resampling and uses a time-varying objective to balance the original and resampled graph branches.
- Analyzed the model's training dynamics across citation networks, improving on D-GCN for CiteSeer (86.63% to 97.26%) and DBLP (80.24% to 82.02%) while identifying degradation on PubMed as branch weight shifted toward the resampled graph.

Manifold Mixup for Personalized Federated Learning 2022

Georgia Institute of Technology

- Integrated manifold mixup into FedPer to regularize the shared representation layers while preserving client-specific personalization.
- Compared local and aggregated model behavior against unregularized FedPer and identified a failure mode in which mixup degrades performance when client models are aggregated before learning stable local representations.

Agent-Based Simulation of Social Network Adoption 2022

Georgia Institute of Technology

- Developed an agent-based model of platform adoption and abandonment that combined individual preferences, social conformity, and sentiment propagation across multiple network topologies.
- Derived a mean-field approximation and compared it with simulation behavior to study how network structure and local interactions produced global adoption patterns.

Long-Horizon Traffic Flow Prediction with Deep Convolutional Networks 2020

IIT Bombay

- Adapted InceptionNet, DenseNet, VGGNet, and shallow convolutional architectures to model spatiotemporal traffic flow at extended prediction horizons.
- Compared network architectures, optimization methods, and temporal preprocessing strategies, with the strongest deep models substantially outperforming the shallow CNN baseline.

Hybrid Visual Positioning with Semantic Filtering and Semi-Direct Odometry 2020

IIT Bombay

- Designed a visual-positioning framework combining semantic filtering with semi-direct visual odometry to reduce feature-matching outliers and estimate camera motion from photometric consistency.
- Specified a Bayesian depth-mapping module and an evaluation protocol against ORB-SLAM on the KITTI and ETH benchmarks.

Professional Experience

TJKM July 2024–November 2025

Data Scientist/Travel Demand Modeler San Jose, CA

- Reverse-engineered and extended desktop analysis software spanning GISDK, Python, batch scripts, and DBF-based data workflows, tracing failures across application and runtime boundaries.
- Built Python pipelines for numerical analysis and structured report generation, with validation and cross-file reconciliation to replace error-prone manual workflows.

Fehr & Peers January 2023–April 2024

Transportation Engineer/Planner San Jose, CA

- Built reproducible Python workflows for cleaning, joining, validating, and analyzing traffic-count, GPS, and travel-demand datasets used in transportation studies.
- Developed project-specific metrics, visualizations, and quality-control checks, replacing recurring spreadsheet analyses with scripted workflows.

Fehr & Peers May 2022–August 2022

Transportation Planning Intern Washington, DC

- Developed Python and COM-based workflows to automate traffic-simulation experiments, including scenario generation, parameter updates, model execution, and output extraction.
- Supported calibration of driver-behavior parameters by comparing simulated traffic patterns with observed data across controlled experimental scenarios.